

✅ Practice Set – Functions Mastery (Level 1 to 3)

Focus: parameters, arguments, return

Q1. ✅

```
function greet(name) {  
  console.log("Hello " + name);  
}  
greet("Yugant");  
greet();
```

Output kya hoga?

Agar second call me koi argument nahi diya gaya, kya hoga?

Q2. ✅

```
function add(a, b) {  
  return a + b;  
}  
console.log(add(5, 7));  
console.log(add(10));  
Second call me kya output aayega aur kyun?
```

Q3. ✅

```
function introduce(name = "Guest", country = "India") {  
  console.log(`${name} is from ${country}`);  
}  
introduce("Yugant", "Japan");  
introduce("Ravi");  
introduce();
```

Teeno calls ke output likho.

Q4. ✅

```
function sumAll(...nums) {  
  let total = 0;  
  for (let n of nums) total += n;  
  return total;  
}  
  
console.log(sumAll(2, 4, 6));  
console.log(sumAll(5));  
console.log(sumAll());
```

Explain karo ki rest parameter kya kar raha hai.

Level 3: Spread Operator & Combination
Focus: spread aur rest dono

Q5. ✅

```
function multiply(a, b, c) {  
  console.log(a * b * c);  
}
```

```
const arr = [2, 3, 4];
multiply(...arr);
```

Yahan ...arr kya kar raha hai? Output kya hoga?

Q6. 

```
function showDetails(...info) {
  console.log(info);
}

const data = ["Yugant", 21, "Surat"];
showDetails(...data);
```

Step by step explain karo ki ... dono jagah kya kar raha hai.

Level 4: Hoisting (Conceptual Challenge)
Focus: JS engine ka working samajhna

Q7. 

```
sayHi();

function sayHi() {
  console.log("Hello!");
}

const greet = function() {
  console.log("Hi there!");
};

greet();
Yeh code bina error chalega?
Agar greet(); ko upar likh do to kya hoga?
```

Q8. 

```
console.log(sum(2, 3));

function sum(a, b) {
  return a + b;
}

console.log(multiply(2, 3));

const multiply = (a, b) => a * b;
Bonus Challenge (Real-Life Style)
Mix of everything you learned
```

Q9. 

```
function orderSummary(customer = "Guest", ...items) {
  console.log(`Customer: ${customer}`);
  console.log(`Items: ${items.join(", ")}`);
  console.log(`Total items: ${items.length}`);
}

orderSummary("Yugant", "Burger", "Fries", "Coke");
orderSummary("Ravi");
orderSummary();
```

practice.txt

Output likho aur samjhao kya ho raha hai har call me.

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✓ Q1: Ek function printAll banao jo jitne bhi arguments aaye, unko line by line print kare.

✓ Q2: Ek function addAll banao jo jitne bhi numbers aaye unka sum return kare.

✓ Q3: concatStrings("Hello", "Yugant", "Singh") → "Hello Yugant Singh"

✓ Q4: findMax(10, 25, 8, 99, 32) → 99

✓ Q5: Ek function separateTypes(...args) banao jo mixed arguments lega (numbers, strings, booleans).

Phir unko unke type ke hisaab se alag-alag arrays me rakhe.

Aur ek object return kare jisme teen arrays ho:

- * numbers → saare numbers
- * strings → saare strings
- * booleans → saare booleans

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✓ Q6: findMin

Jitne bhi numbers aaye, unme se minimum return kare.

findMin(10, 25, 8, 99, 32); // 8

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✓ Q8: filterByType ==> Ek function filterByType(type, ...args) banao.

Pehla argument hoga type → "number", "string", ya "boolean".

Baaki jitne arguments aaye, unme se sirf wahi values return karni hain jo type ke match karte ho.

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Q9: sumEvenOdd

Goal:

Function me numbers pass honge → ...nums

Hume calculate karna hai:

even numbers ka sum

odd numbers ka sum

Return ek object jisme { even: sumEven, odd: sumOdd }

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 Easy Higher-Order Function Practice

✓ Q1 - Function as Argument

Ek function greetUser likho jo ek callback function lega aur usko run karega.

✓ Q2 - What is the difference between function statement and function expression in terms of hoisting :

Ans- function statement can be hoisted ,means they can be call before the define but function expression can not hold the hoisting property

✓Q3 - convert the given function in to arrow function.

```
function multiply(a,b){  
    return a*b;  
}
```


✓Q4 - How many Argument pass in the function

```
function demo(a,b,c){}  
  
demo(12,3);
```


✓Q5 - Predict the output of the given code

```
function sayHii(name = "Guest"){  
    console.log(name);  
}  
sayHii();
```


✓Q6 -What does the ... operator mean in the parameter

✓Q7 - Use the rest parameter to accept any number of score and return the sum of score

✓Q8 - Fixed the function with early return

```
function checkAge(age){  
  
    if(age>18){  
        console.log("You can vote");  
    } else {  
        console.log("Its too early");  
    }  
}  
checkAge(12);
```


✓Q9 - What does it mean when we say that "Function are first class citizens"

✓Q10 - Pass an function into another function and execute it

✓Q11 - What is Higher order function

✓Q12 -What is Pure and Impure function in js

✓Q13 What is closure and when is created ?

✓Q14 - Convert this this normal function into IIFE function

```
function init(){  
    console.log("initialised");  
}
```

```
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[✓] Higher Order Function + Closure + Currying level
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[✓] Q1. (Function as Parameter)

Ek HOF runTwice banao jo ek function accept kare aur usko 2 baar run kare.
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[✓] Q2. (Closure Basic)

Ek function counter banao jo ek function return kare. Har baar call karne par
counter +1 ho jaye.
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◆ Medium Level
[✓] Q3. (Currying Sum)

Ek function sum banao jo 3 arguments ko alag alag function calls me accept kare
aur unka sum return kare.
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[✓] Q4. Greater Than with Closure

Ek function greaterThan(n) banao jo ek function return kare
Ye inner function ek number lega aur check karega ki wo n se bada hai ya nahi.
Agar bada hai → true, nahi → false

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[✓] Mini Practice Set (Self Try)

[✓] Q6: Ek function repeatN banao jo ek function aur number n accept kare.
Function ko n baar call kare.

[✓] Q7: Ek function multiplyBy(x) banao jo ek function return kare. Inner
function kisi number ko x se multiply kare.

[✓] Q8: Ek function isBetween(min, max) banao jo number accept kare aur check
kare ki number min aur max ke beech hai ya nahi.

[✓] Q9: Ek function chainSum(a) banao jo multiple chained calls ke through sum
calculate kare. Example: chainSum(1)(2)(3)(4)() → 10.

[✓] Q10: Ek function mapArray(arr, func) banao jo array aur function accept kare
aur new array return kare jisme function applied ho.

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```